

Deliverable 1: Project Specification

Team Composition		
José Nunes	João Lourenço	Leonor Afonso
73137	72904	73491

Project Title: Behavioural Segmentation of Travel Agency and Tour Operator Bookings in the Hotel/Resort Industry

Problem Statement

The hospitality industry increasingly relies on data-driven strategies to optimise revenue and manage relationships with distribution intermediaries. Travel agencies (TA) and tour operators (TO) are significant booking intermediaries, yet their behaviours are heterogeneous and not directly observable through any single variable. This project investigates whether latent behavioural clusters exist among TA/TO-mediated bookings by jointly analysing distribution channel, lead time, and booking amendments, with the goal of helping hotels better understand intermediary profiles and support decisions around channel management, pricing, and inventory planning. With 119,390 bookings across two Portuguese hotels (2015–2017), the dataset provides sufficient scale and diversity to reveal meaningful latent structure, yielding actionable insights for channel optimisation and operational planning. insights for channel optimisation, commission structuring, and operational planning.

Problem Framing (RQ1)

RQ1: Are there hidden behavioural clusters among travel agency and tour operator bookings when jointly analysing distribution channel, lead time, and volume of booking modifications?

What do we want to segment?

The unit of analysis is the individual hotel booking record. The target sub-population is bookings made through TA/TO channels, where intermediary behaviour (rather than direct guest behaviour) shapes the booking profile. We seek groups of bookings that share coherent patterns along three dimensions: (i) the specific distribution sub-channel used; (ii) how far in advance the booking was made; and (iii) how many times the booking was subsequently modified.

What would a useful segmentation reveal?

Meaningful clusters would correspond to distinct intermediary booking strategies, such as:

- Last-minute, low-modification bookings from direct TA channels - suggesting opportunistic placements;
- Long-horizon, high-modification bookings - suggesting contract-based operators that negotiate iteratively;
- Stable, far-in-advance bookings from GDS or corporate channels - suggesting structured corporate travel.

These profiles carry direct managerial implications for pricing flexibility, overbooking policies, and partner incentive structures. Post-hoc profiling with outcome variables (e.g., cancellation rate, ADR) will further enrich interpretation without leaking into the clustering itself.

Why is clustering appropriate here?

Clustering is the methodologically appropriate paradigm for this problem on three mutually reinforcing grounds: the absence of labelled outcomes, the nature of the research objective, and the structure of the feature space.

First, the problem is inherently unsupervised. No ground-truth labels exist for intermediary behavioural types. Without a pre-defined target variable, supervised learning is inapplicable. The task is fundamentally one of discovery: we seek to determine whether coherent latent groupings exist, not to predict a known outcome for new observations.

Second, clustering partitions the observation space into regions of high intra-group similarity and high inter-group dissimilarity, allowing distinct booking profiles to surface empirically. This makes clustering the appropriate tool for generating and substantiating the hypothesis that meaningful intermediary archetypes exist.

Third, the feature space is jointly informative. A booking's distribution sub-channel, lead time, and number of amendments are individually ambiguous. It is the combination of these dimensions that identifies a coherent booking strategy. Clustering operates directly on this joint representation, making it better suited than any univariate analysis or sequential stratification.

Finally, the scale and composition of the dataset make clustering statistically viable. With 119,390 booking records spanning two hotel properties and three years, the sub-population of TA/TO bookings is sufficiently large to support robust parameter estimation across multiple clusters, while the temporal and property-level diversity reduces the risk that any discovered structure is an artefact of a single hotel's operational patterns. Taken together, these three grounds, constitute a principled and sufficient justification for adopting a clustering approach.

Representation / Preprocessing Choices (RQ2)

RQ2: Which preprocessing and encoding choices are justified for this problem, and how might they affect similarity and clustering?

Feature Selection and Leakage Control

The clustering feature set is constructed at index time to ensure no outcome information leaks into the representation.

Primary variables (directly motivated by RQ1):

- **distribution_channel:** categorical; captures the TA/TO sub-channel structure.
- **lead_time:** numerical; measures the planning horizon of the booking.
- **booking_changes:** integer; quantifies post-confirmation amendment behaviour.

Contextual enrichment variables (to improve cluster separability and interpretability):

- **market_segment:** complements distribution_channel, providing finer intermediary granularity.
- **deposit_type:** signals commitment and risk profile of the booking.
- **customer_type:** distinguishes transient, contract, and group bookings, relevant to intermediary type.
- **stays_in_week_nights + stays_in_weekend_nights:** booking length as a proxy for travel purpose.
- **adr (Average Daily Rate):** pricing context; used only if available at booking time.

Excluded Variables and Justifications

- **is_canceled:** post-event outcome; not available at index time.
- **reservation_status / date:** post-arrival status; directly encodes the outcome.
- **agent / company:** Exclude (high cardinality): identifier-like fields with 100s of unique values; cannot be responsibly encoded without creating noise dimensions.

Encoding and Scaling Strategy

The mixed-type dataset requires a unified Euclidean representation via scikit-learn's ColumnTransformer. Categorical variables are one-hot encoded with rare categories collapsed into "Other", while numerical variables are standardised with StandardScaler — with RobustScaler tested as a robustness variant given known outliers. The inclusion of booking_changes is justified since PMS data was captured the day prior to arrival, making amendments observable at index time. All exclusions are grounded in leakage or cardinality concerns.

Limitation, Risk, or Ethical Issues

Here are some methodological and ethical considerations that apply to this study:

- **Temporal scope:** the dataset covers 2015-2017; booking behaviours may have evolved significantly since then. Findings should not be generalised to current market conditions without validation.
- **Leakage boundary for booking_changes:** while we justify its inclusion, any post-hoc interpretation must acknowledge that the exact number of changes observed may be influenced by the type of intermediary, creating a risk of circular reasoning. We will flag this in the final report.
- **Representativeness:** the data covers only two Portuguese hotels. Cluster profiles discovered here may not generalise to other markets, hotel categories or geographic contexts.